The Citric Acid Cycle



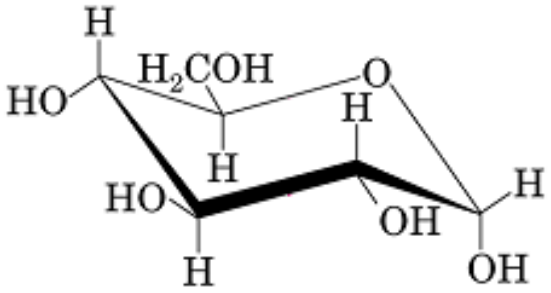
Hans Krebs, 1900–1981

The Citric Acid Cycle

Learning goals:

- Cellular respiration
- Conversion of pyruvate to activated acetate
- Reactions of the citric acid cycle
- Regulation of the citric acid cycle
- Amphibolic nature of citric acid cycle intermediates
- Mechanisms of replenishing citric acid cycle intermediates

Only a Small Amount of Energy Available in Glucose Is Captured in Glycolysis

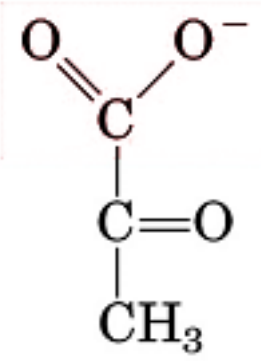


GLUCOSE

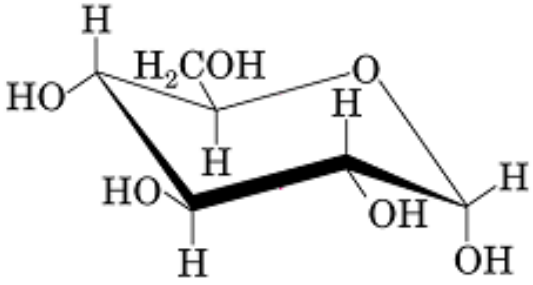
Glycolysis

$\Delta G'^{\circ} = -146$ kJ/mol

2



Pyruvate



Full oxidation (+ 6 O_2)

$\Delta G'^{\circ} = -2{,}840$ kJ/mol

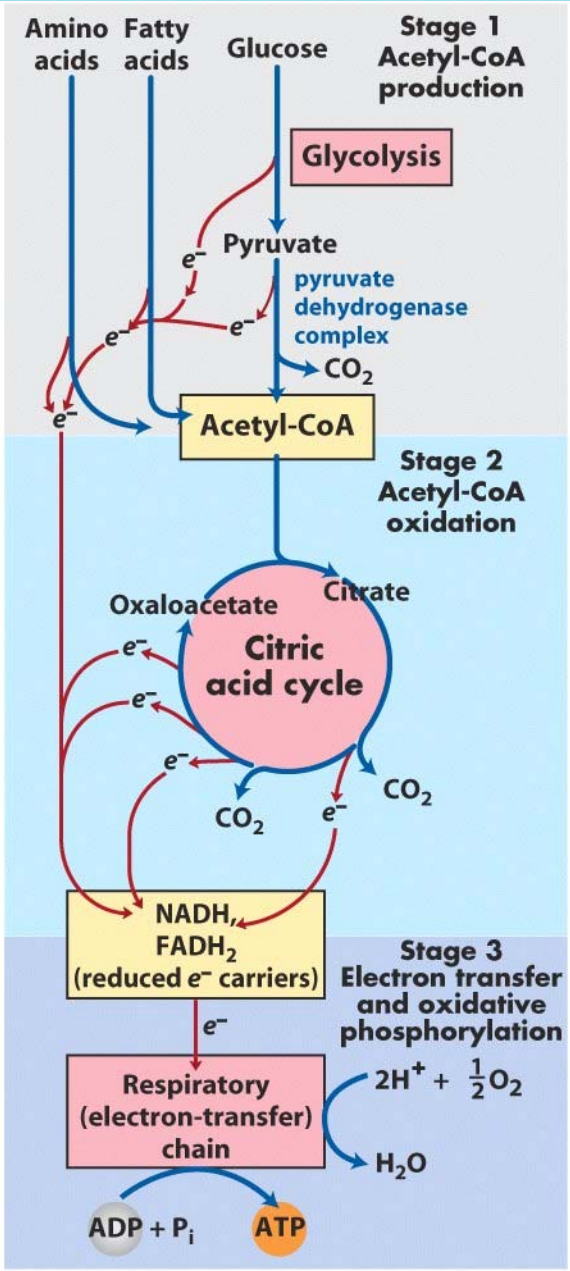
$6\ CO_2 + 6\ H_2O$

Cellular Respiration

- Process in which cells consume O_2 and produce CO_2
- Provides more energy (ATP) from glucose than glycolysis
- Also captures energy stored in lipids and amino acids
- Evolutionary origin: developed about 2.5 billion years ago
- Used by animals, plants, and many microorganisms
- Occurs in three major stages:
  - acetyl CoA production
  - acetyl CoA oxidation
  - electron transfer and oxidative phosphorylation

Cellular Respiration

Cellular respiration is the set of metabolic reactions and processes that take place in the cells of organisms to convert biochemical energy from nutrients into adenosine triphosphate (ATP)



Respiration: Stage 1
Acetyl-CoA Production

Generates some ATP, NADH, $FADH_2$

Carbohydrates release 1/3 of total potential CO_2 during Stage 1.

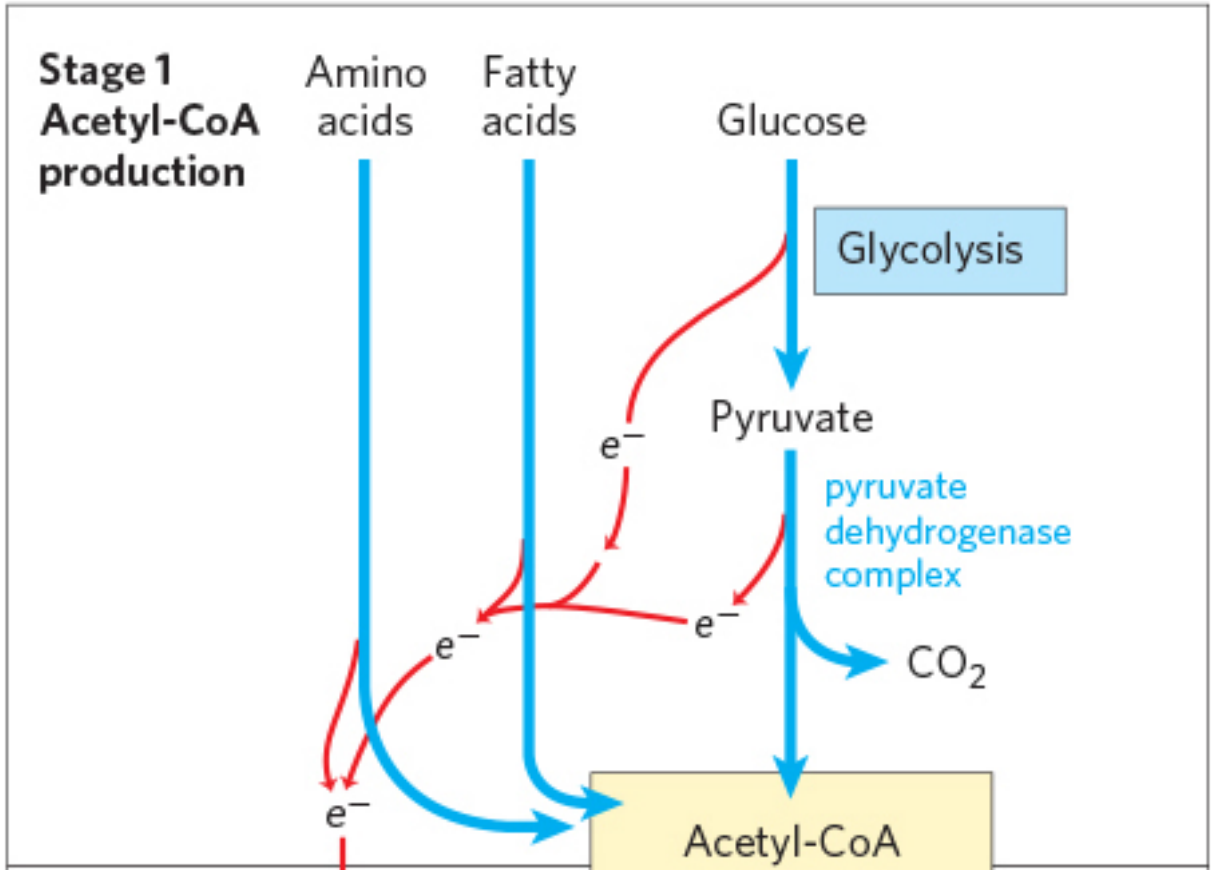


Figure 16.1
Nelson/Cox, *Lehninger Principles of Biochemistry*, 7e, © 2017
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Respiration: Stage 2
Acetyl-CoA Oxidation

Generates more NADH, $FADH_2$, and one GTP

Remaining carbon atoms from carbohydrates, amino acids, and fatty acids are released during Stage 2.

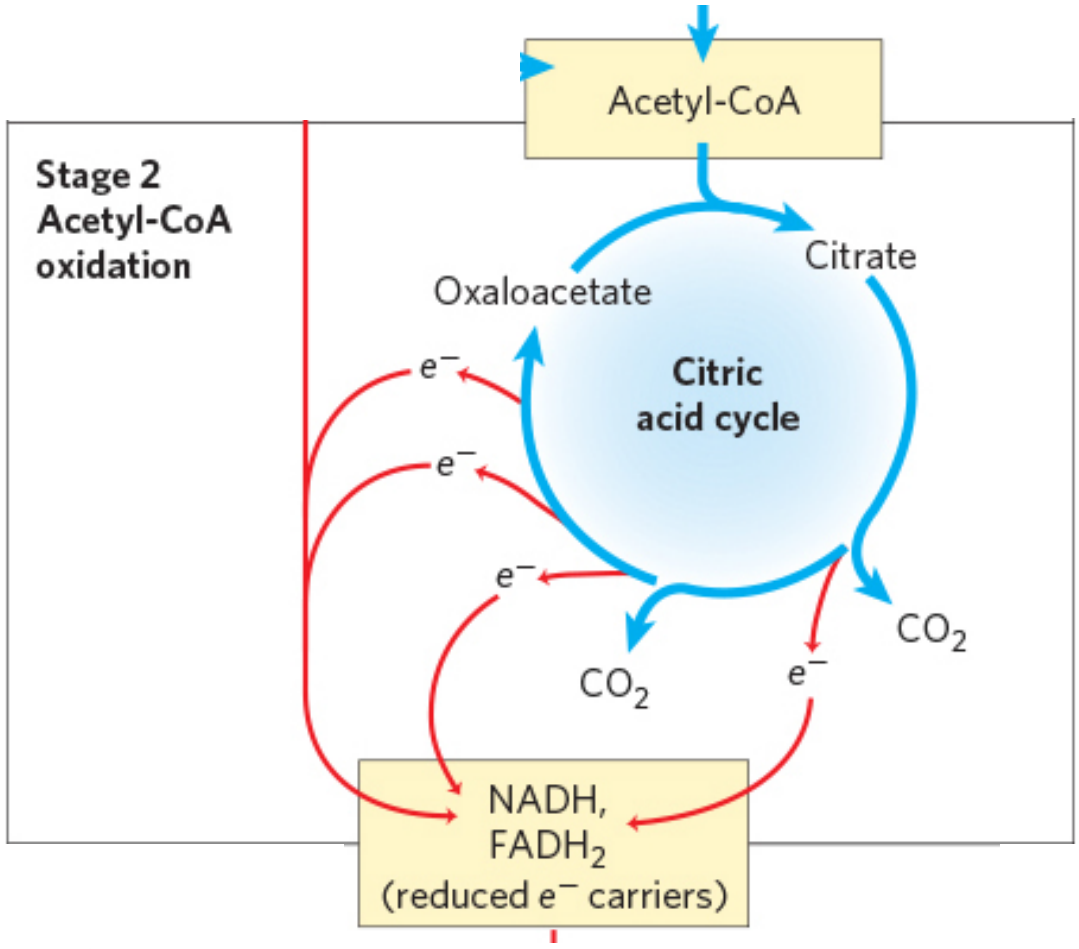


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Respiration: Stage 3
Oxidative Phosphorylation

Generates the vast majority of ATP during catabolism

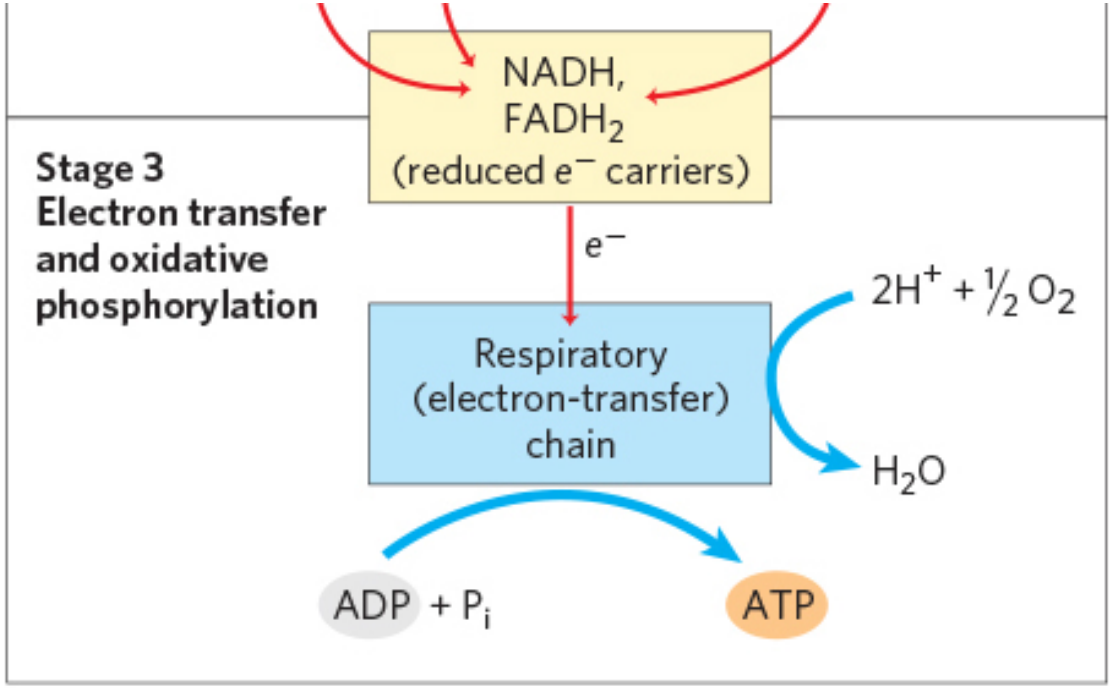


Figure 16.1
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In Eukaryotes, Stages 2 and 3 Are Localized to the Mitochondria

- Glycolysis occurs in the cytoplasm.
- Citric acid cycle occurs in the mitochondrial matrix†.
- Oxidative phosphorylation occurs in the inner membrane.

†Except succinate dehydrogenase, which is located in the inner membrane

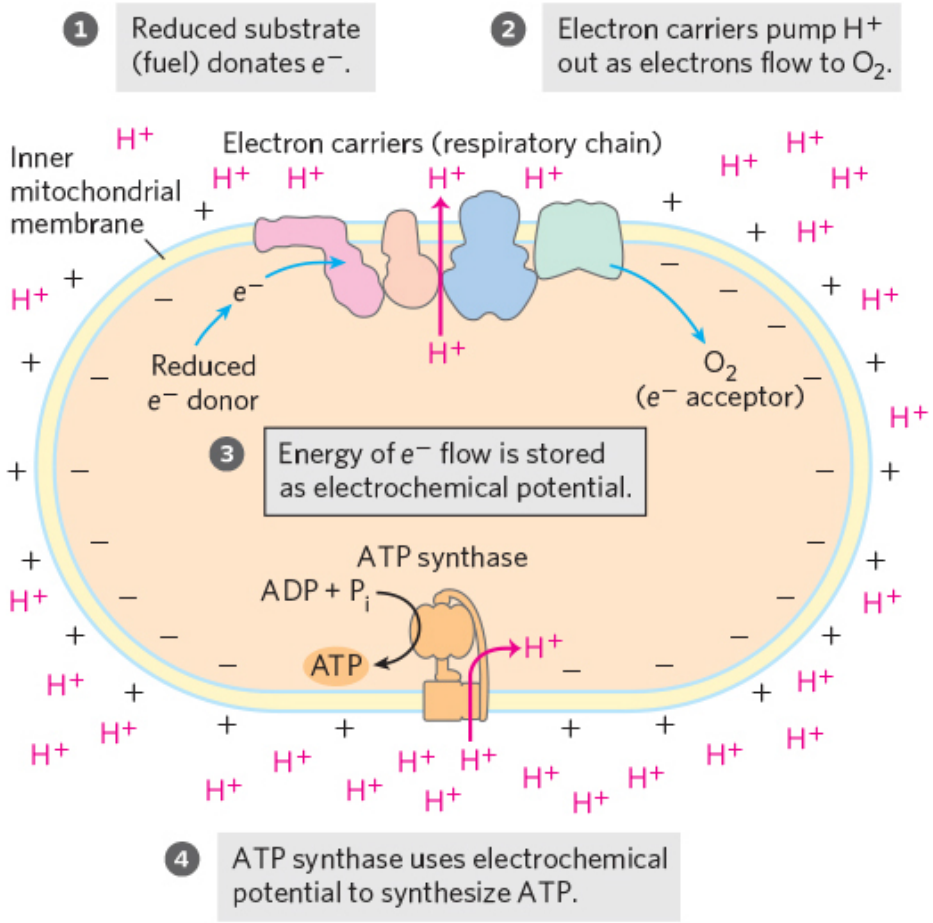


Figure 19.1
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Conversion of Pyruvate to Acetyl-CoA

- Net reaction:
  - oxidative decarboxylation of pyruvate
  - first carbons of glucose to be fully oxidized
- Catalyzed by the pyruvate dehydrogenase complex
  - requires 5 coenzymes
  - TPP, lipoyllysine, and FAD are prosthetic groups.
  - NAD^+ and CoA-SH are co-substrates.

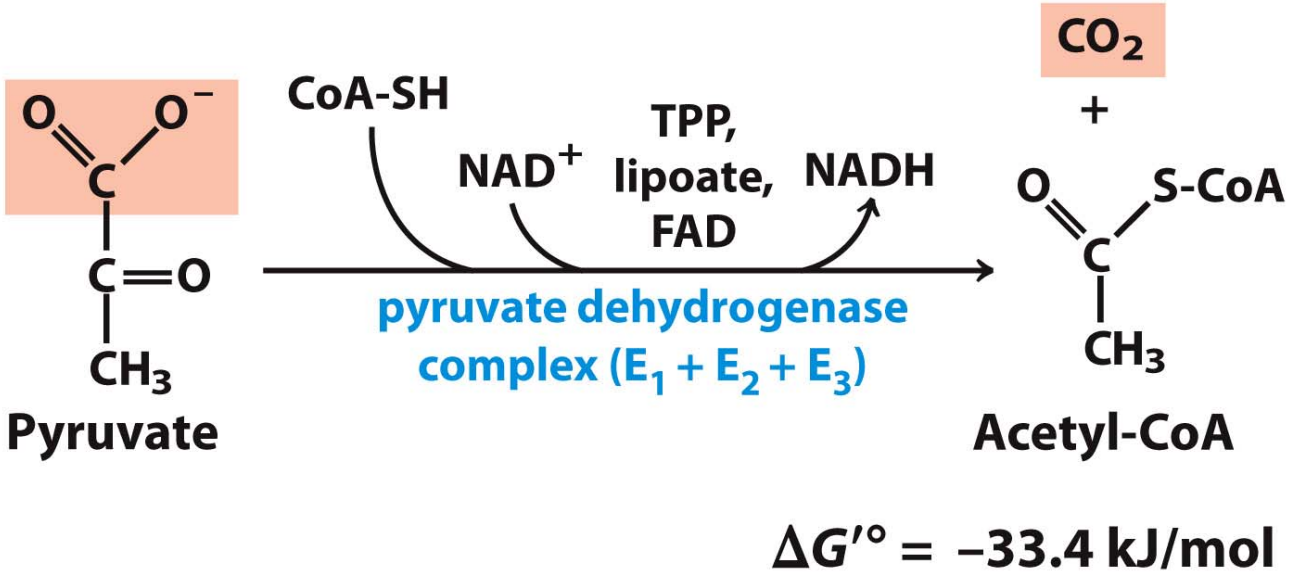


Figure 16-2
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Structure of Coenzyme A

- Coenzymes are not a permanent part of the enzymes' structure.
  - They associate, fulfill a function, and dissociate.
- The function of CoA is to accept and carry acetyl groups.

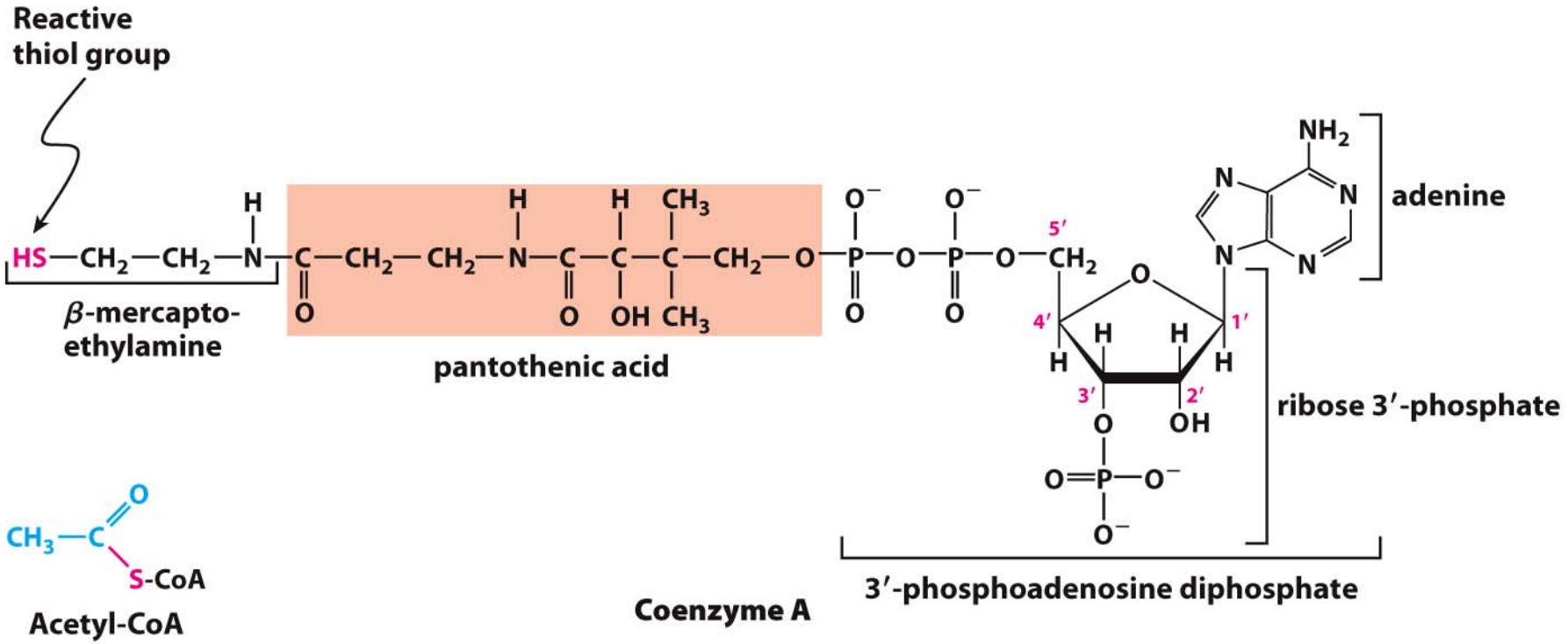


Figure 16-3
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Structure of Lipoyllysine

- Prosthetic groups are strongly bound to the protein.
  - The lipoic acid is covalently linked to the enzyme via a lysine residue

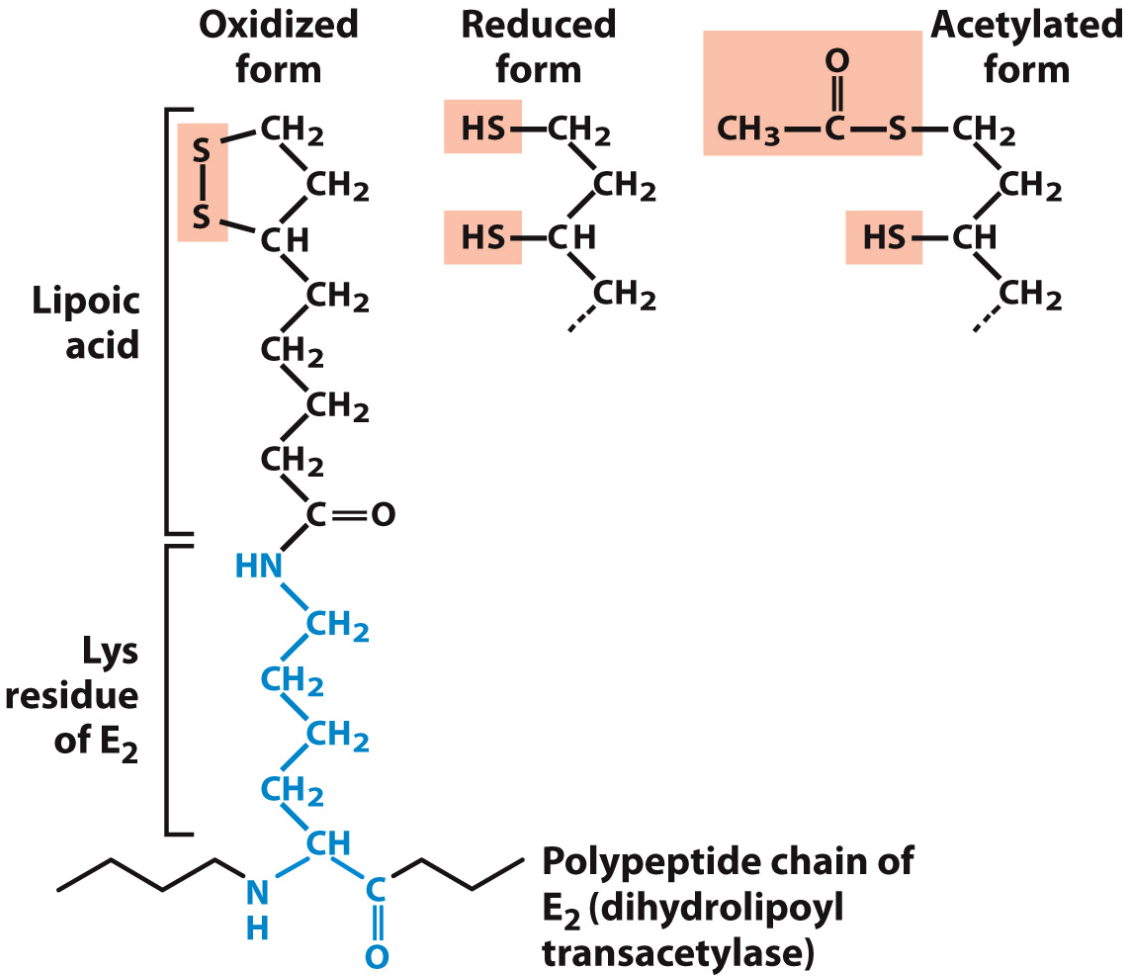


Figure 16-4
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Pyruvate Dehydrogenase Complex (PDC)

- PDC is a large (up to **10 MDa**) multienzyme complex.
  - pyruvate dehydrogenase (E_1)
  - dihydrolipoyl transacetylase (E_2)
  - dihydrolipoyl dehydrogenase (E_3)

- Advantages of multienzyme complexes:
  - The short distance between catalytic sites allows channeling of substrates from one catalytic site to another.
  - Channeling minimizes side reactions.
  - The regulation of activity of one subunit affects the entire complex.

Overall Reaction of PDC

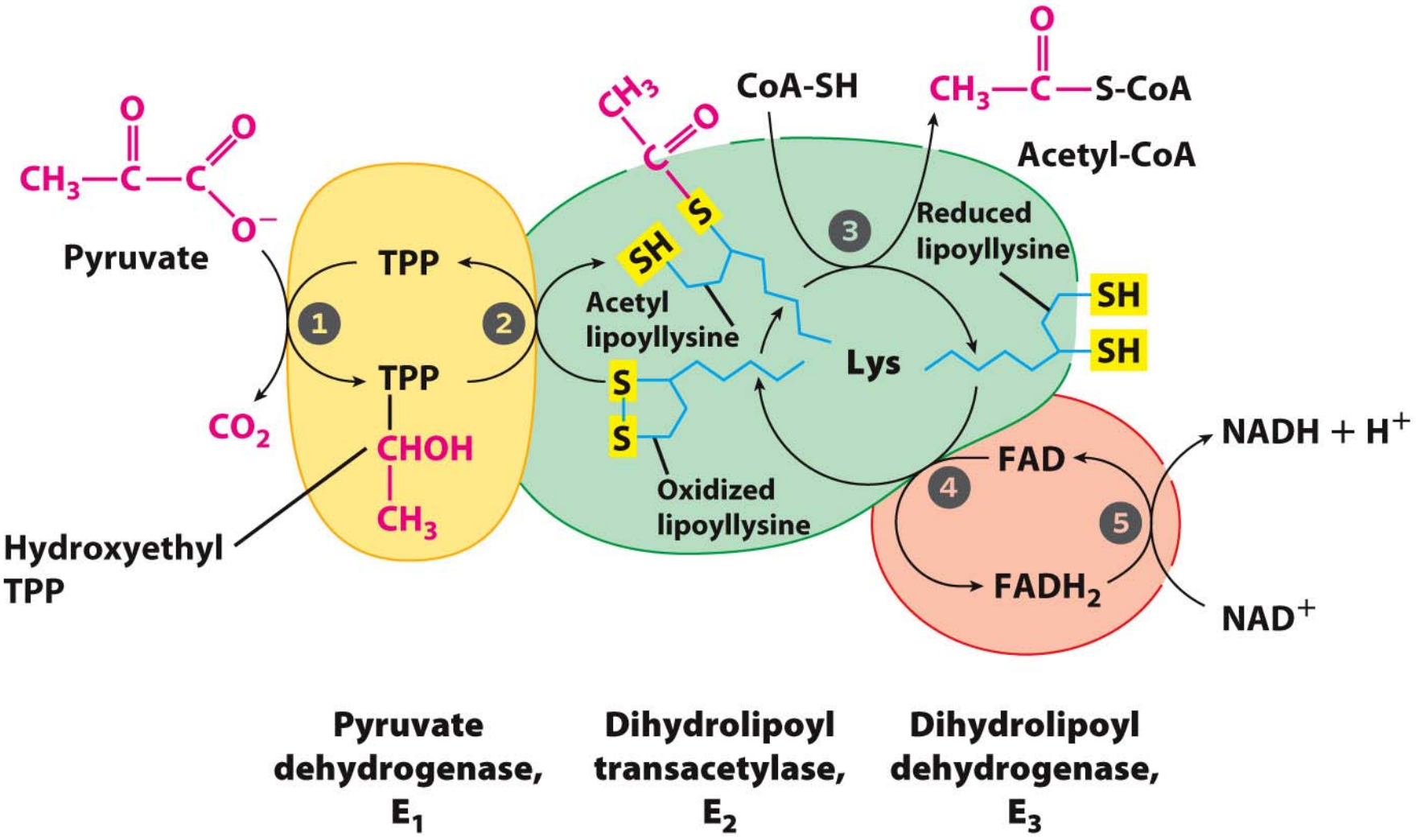
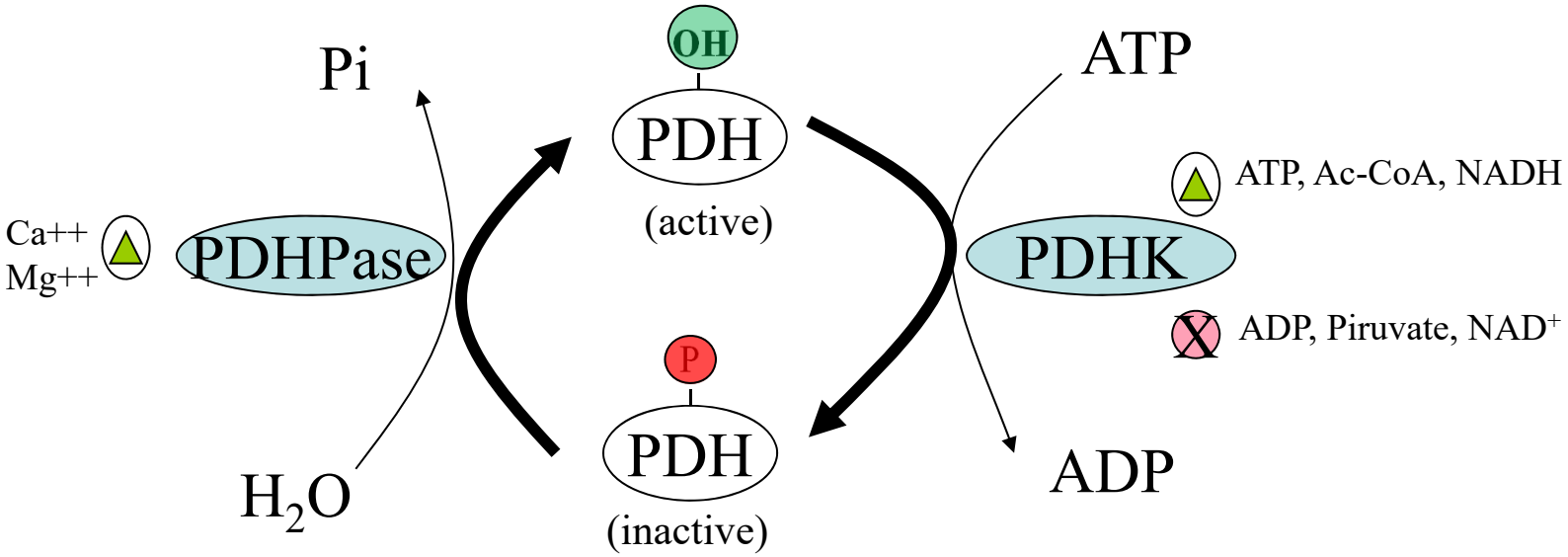


Figure 16-6
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Covalent regulation of PDH complex



The Citric Acid Cycle (CAC)

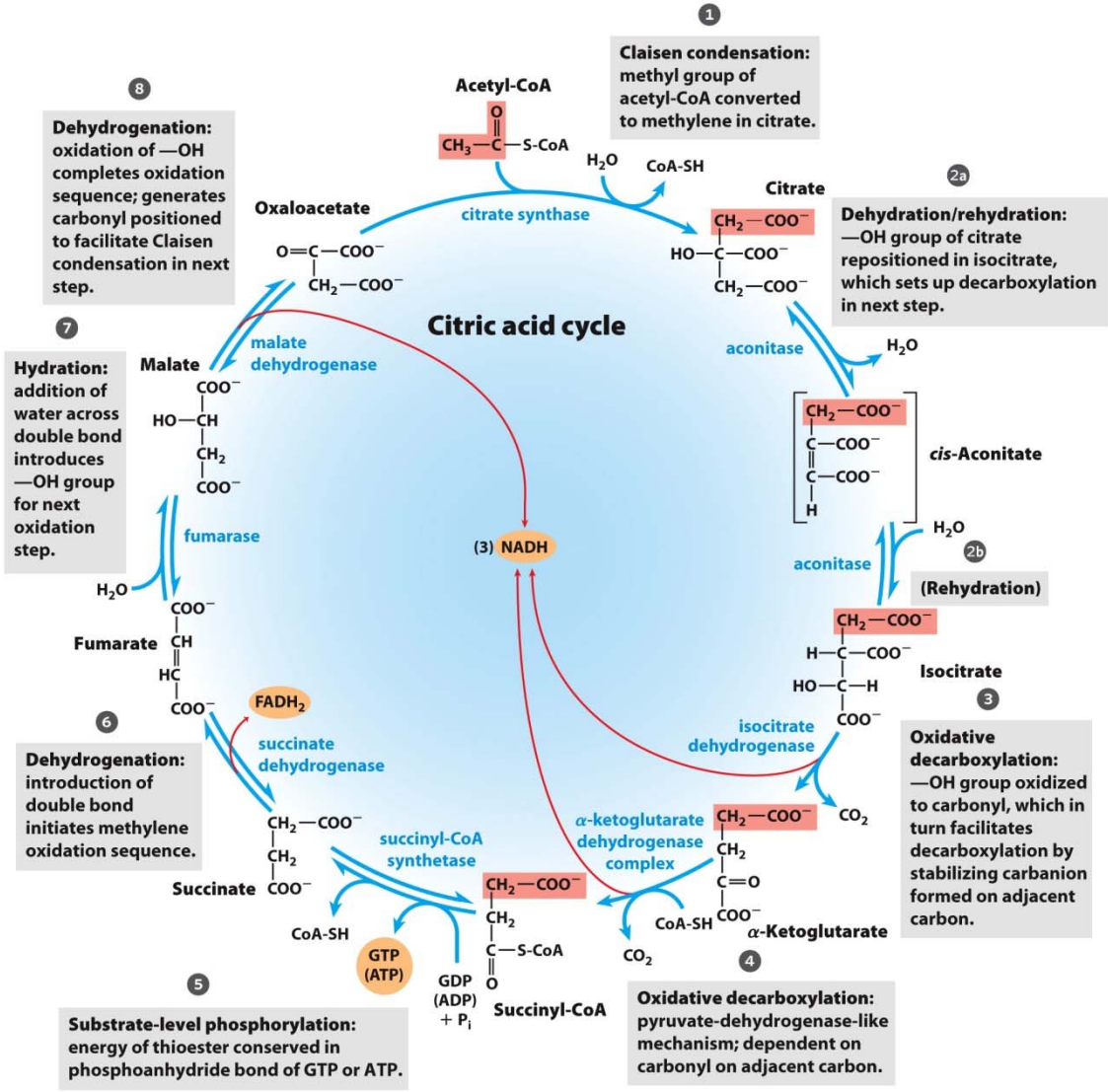


Figure 16-7

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CAC Intermediates Are Amphibolic

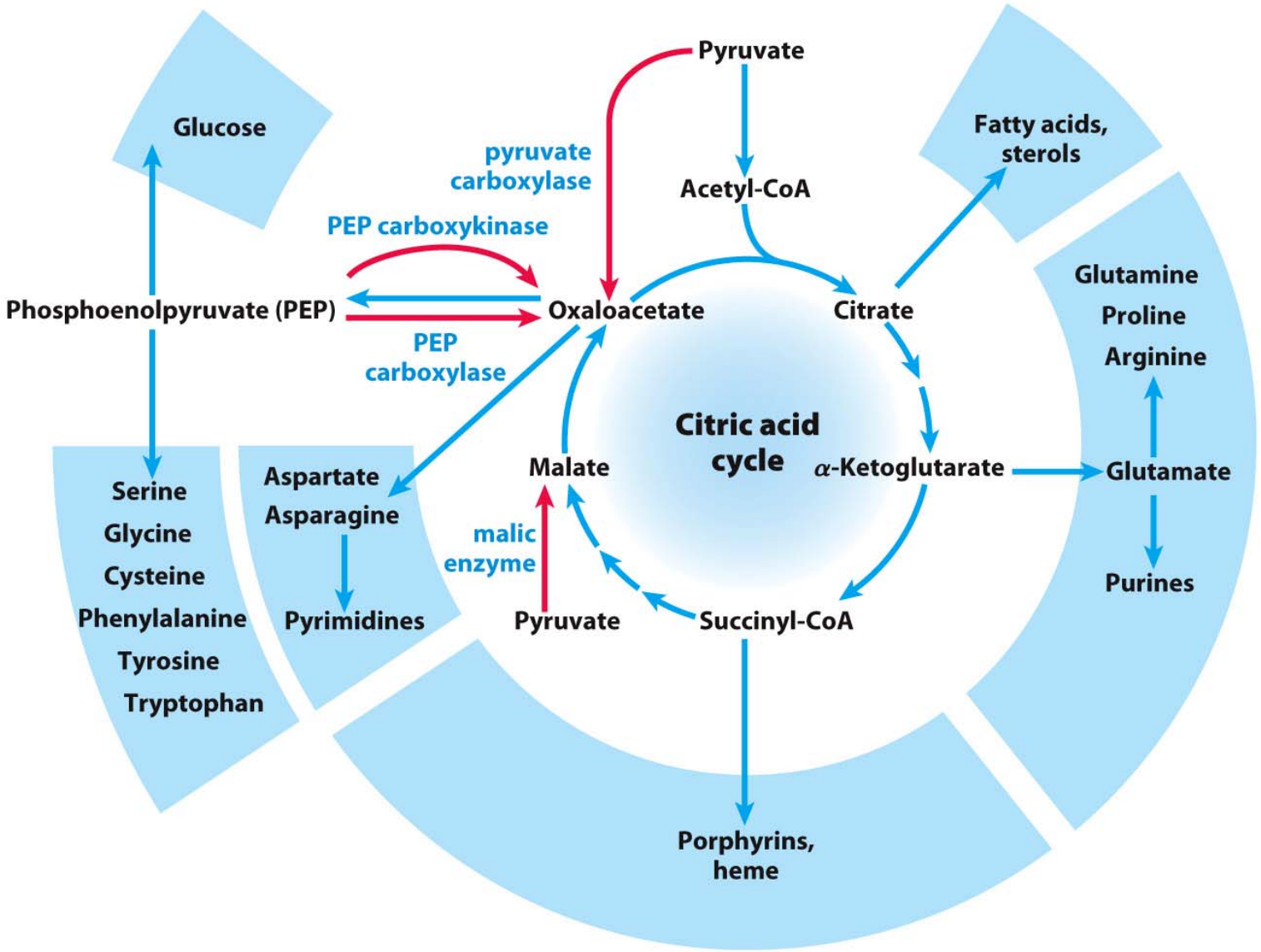
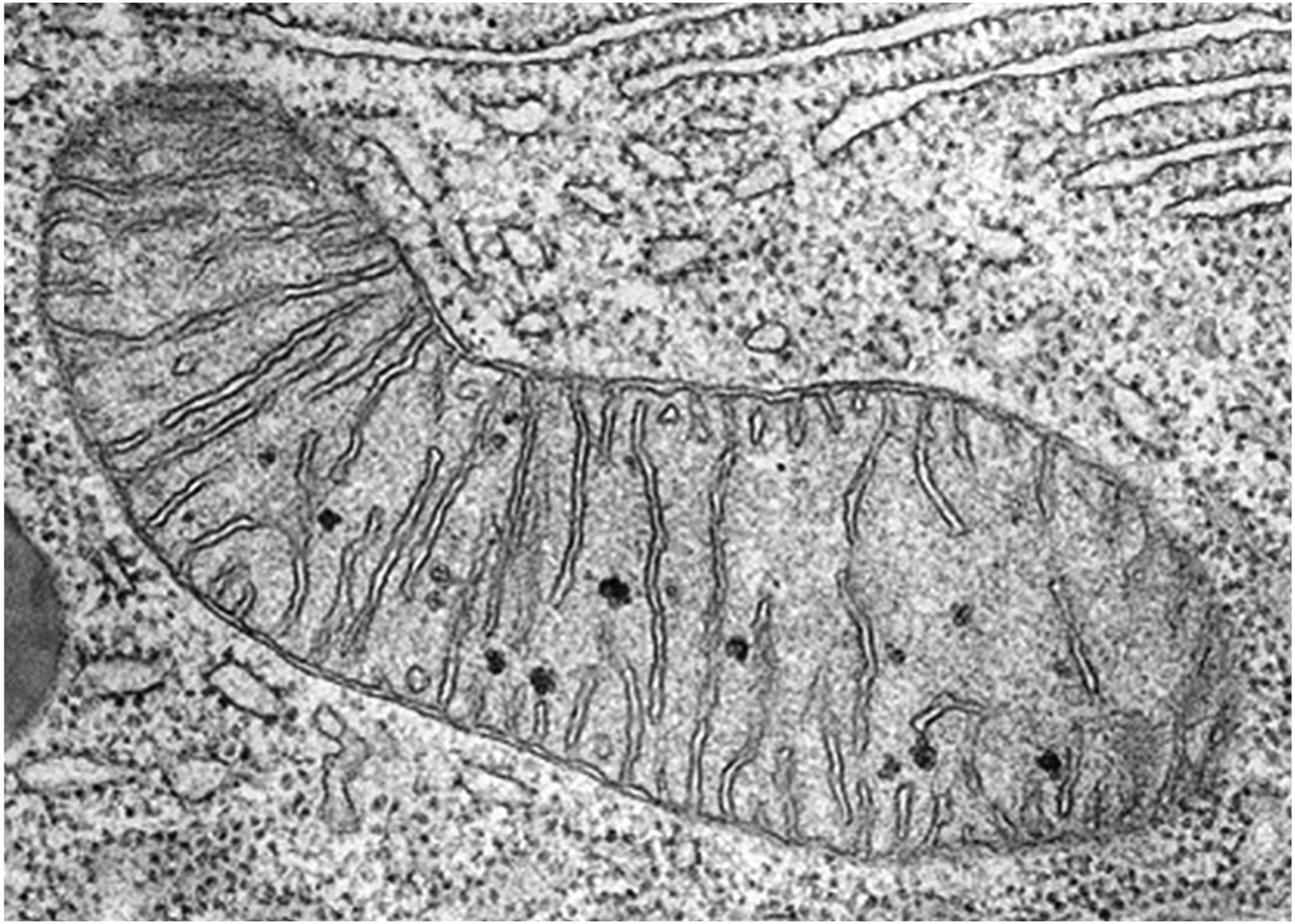
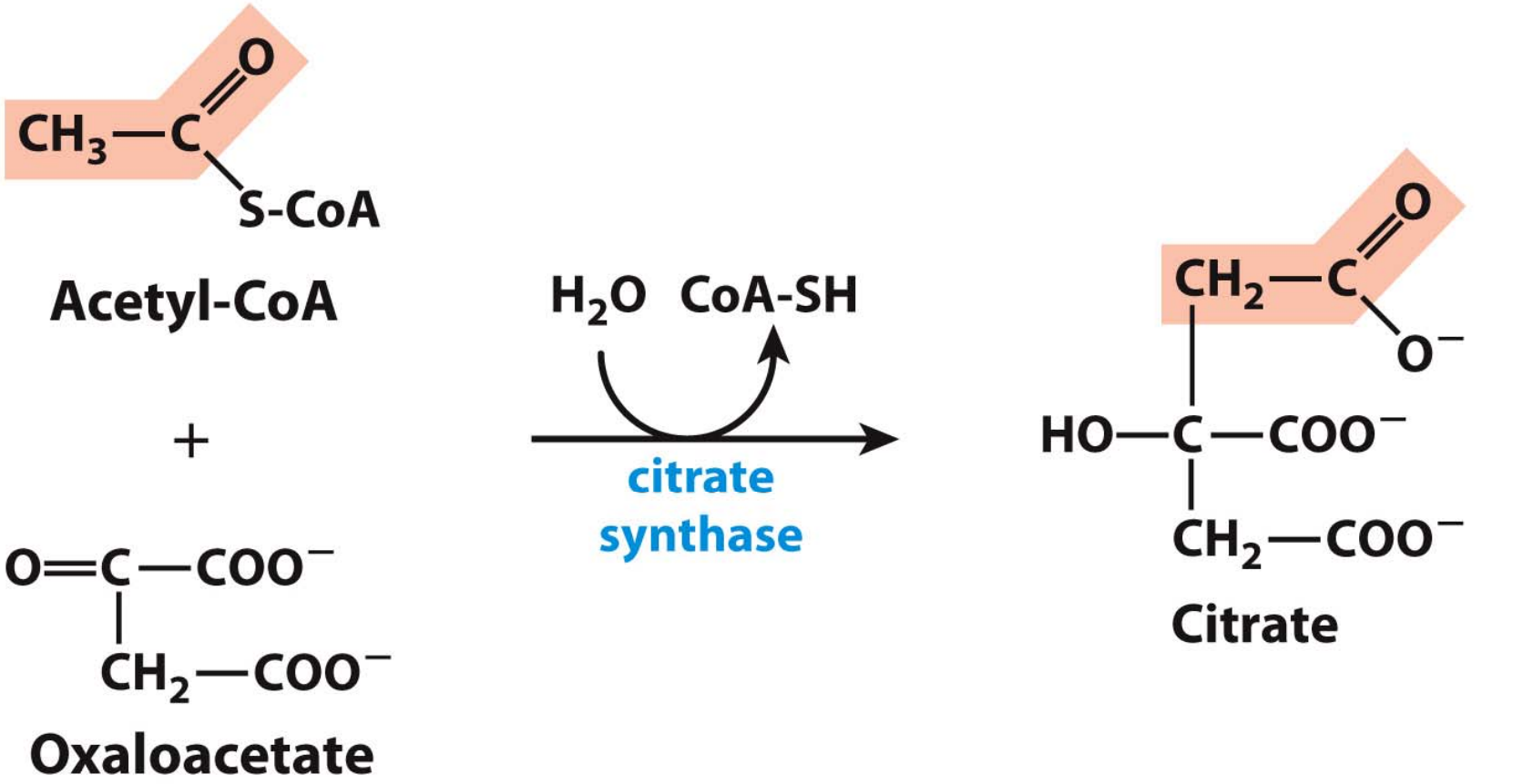


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The mitochondria: the scenary



C-C Bond Formation by Condensation of Acetyl-CoA and Oxaloacetate



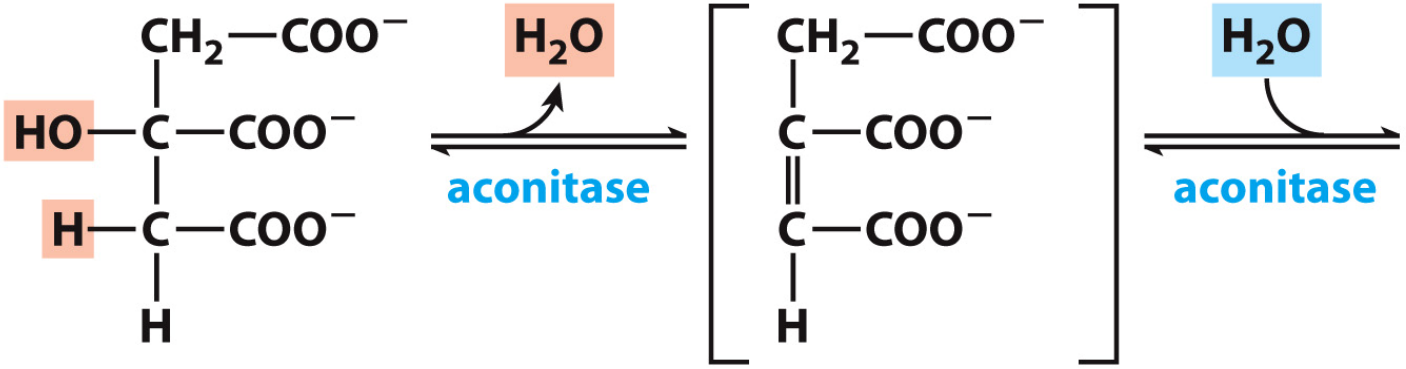
$\Delta G'° = -32.2$ kJ/mol

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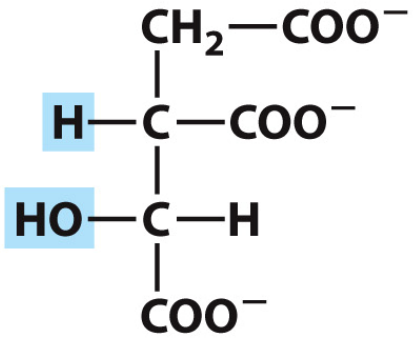
Citrate Synthase

- Condensation of acetyl-CoA and oxaloacetate
- The only reaction with C-C bond formation
- Uses acid/base catalysis
  - Carbonyl of oxaloacetate is a good electrophile.
  - Methyl of acetyl-CoA is not a good nucleophile…
  - …unless activated by deprotonation.
- Rate-limiting step of CAC
- Activity largely depends on [oxaloacetate].
- Highly thermodynamically favorable/irreversible
  - regulated by substrate availability and product inhibition

Isomerization by Dehydration/Rehydration



Citrate **_cis_-Aconitate**



Isocitrate

$\Delta G'° = 13.3$ kJ/mol

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Oxidative Decarboxylation by Isocitrate Dehydrogenase

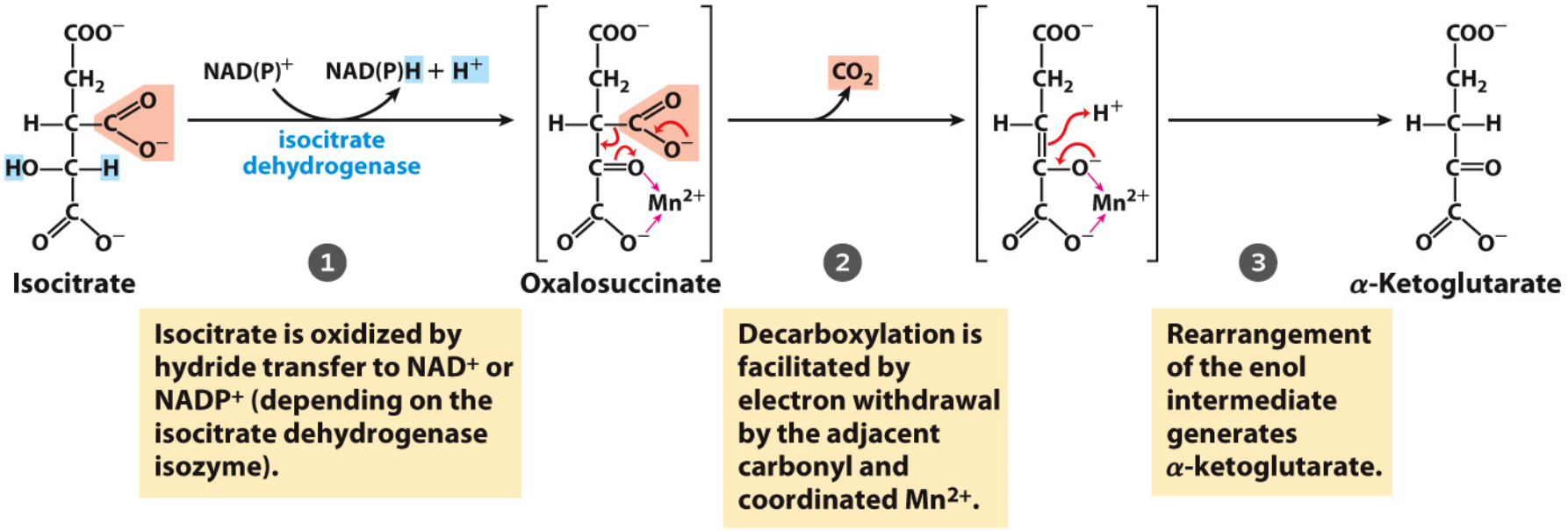
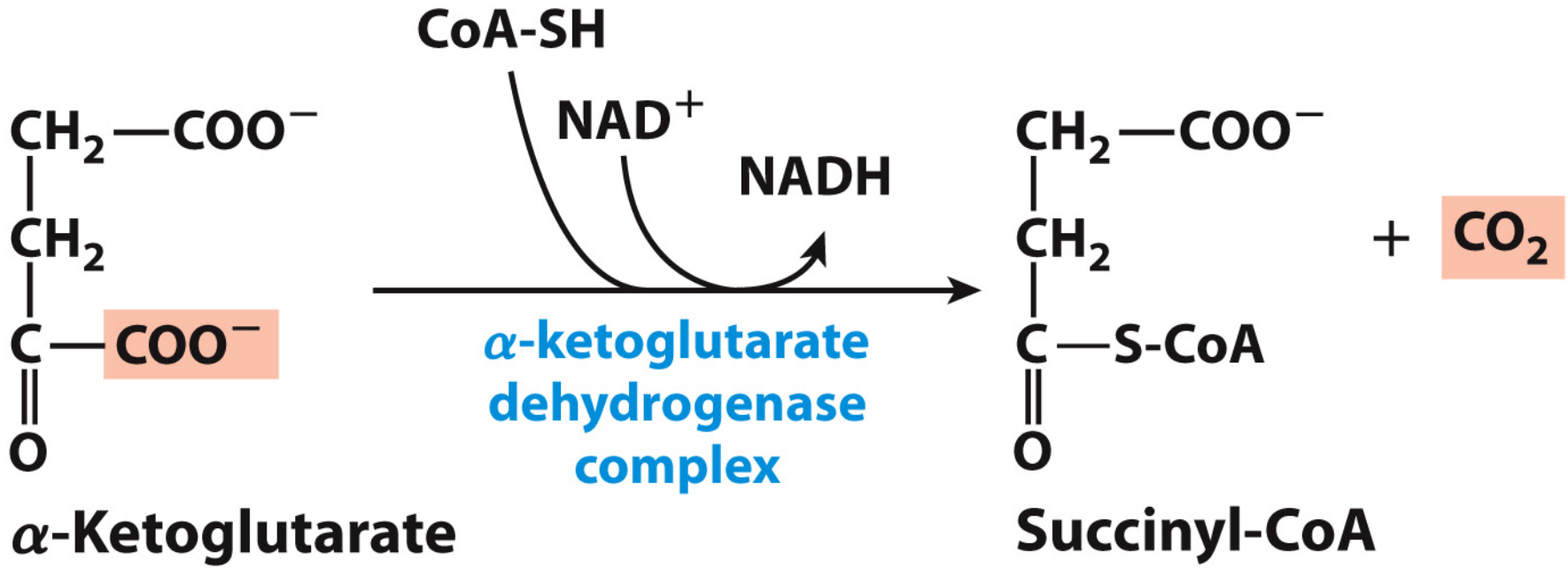


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- Isozymes are specific for $NADP^+$ (cytosolic) or NAD^+ (mitochondrial).
- Decarboxylation releases CO_2.
- Highly favorable/irreversible and regulated by [ATP]

Final Oxidative Decarboxylation by α-Ketoglutarate Dehydrogenase



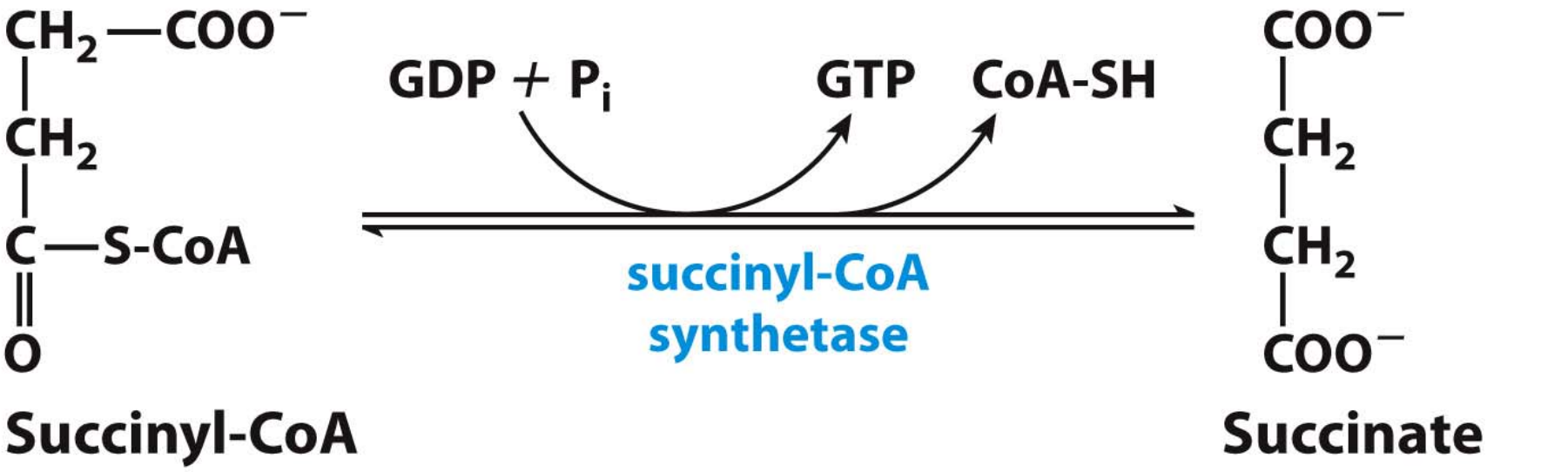
$\Delta G'^{\circ} = -33.5$ kJ/mol

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α-Ketoglutarate Dehydrogenase

- Last oxidative decarboxylation
  - – net full oxidation of all carbons of glucose
    - after two turns of the cycle
    - carbons not directly from glucose because carbons lost came from oxaloacetate, not acetate
- Succinyl-CoA is another high-energy thioester bond.
- Highly thermodynamically favorable/irreversible
  - – regulated by product inhibition

Generation of GTP Through Thioester: Substrate-Level Phosphorylation by Succinyl-CoA Synthetase

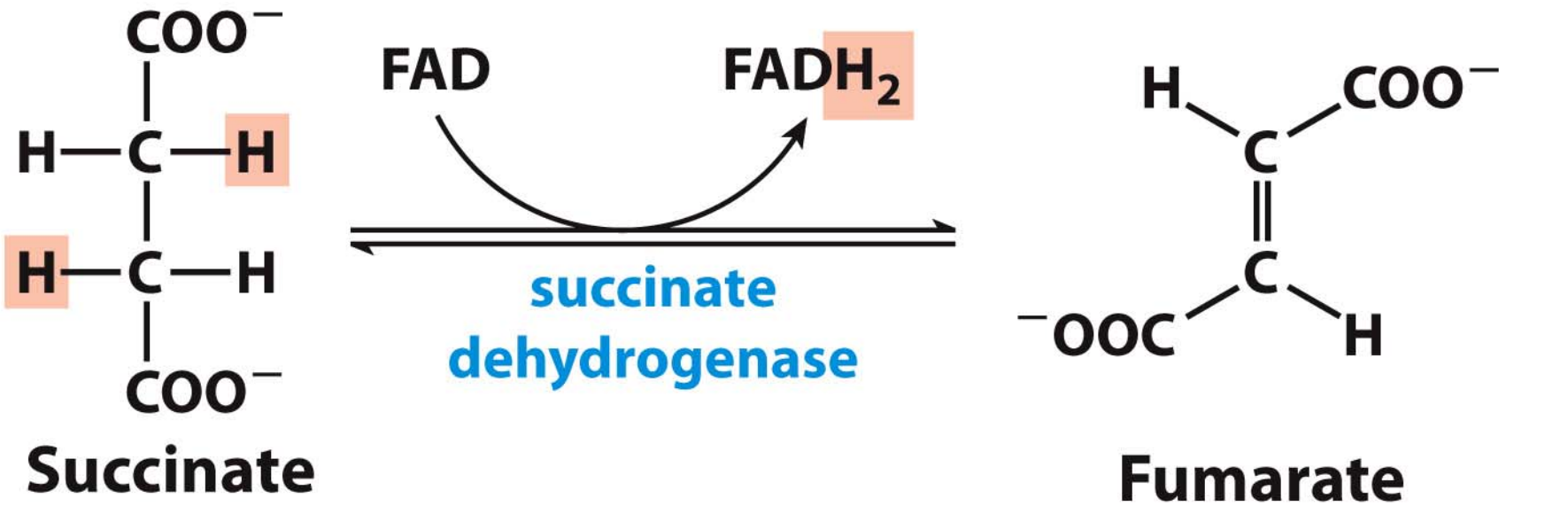


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Succinyl-CoA Synthetase

- Substrate-level phosphorylation
- The energy of thioester allows for incorporation of inorganic phosphate.
- Goes through a phospho-enzyme intermediate
- Produces GTP, which can be converted to ATP
- Slightly thermodynamically favorable/reversible
  - product concentration kept low to pull forward

Oxidation of an Alkane to Alkene by Succinate Dehydrogenase



$\Delta G'° = 0$ kJ/mol

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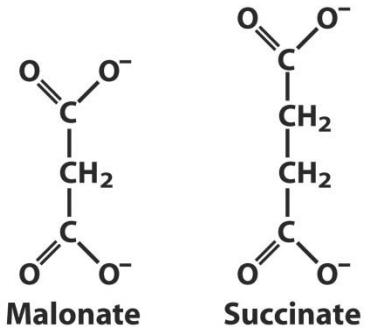
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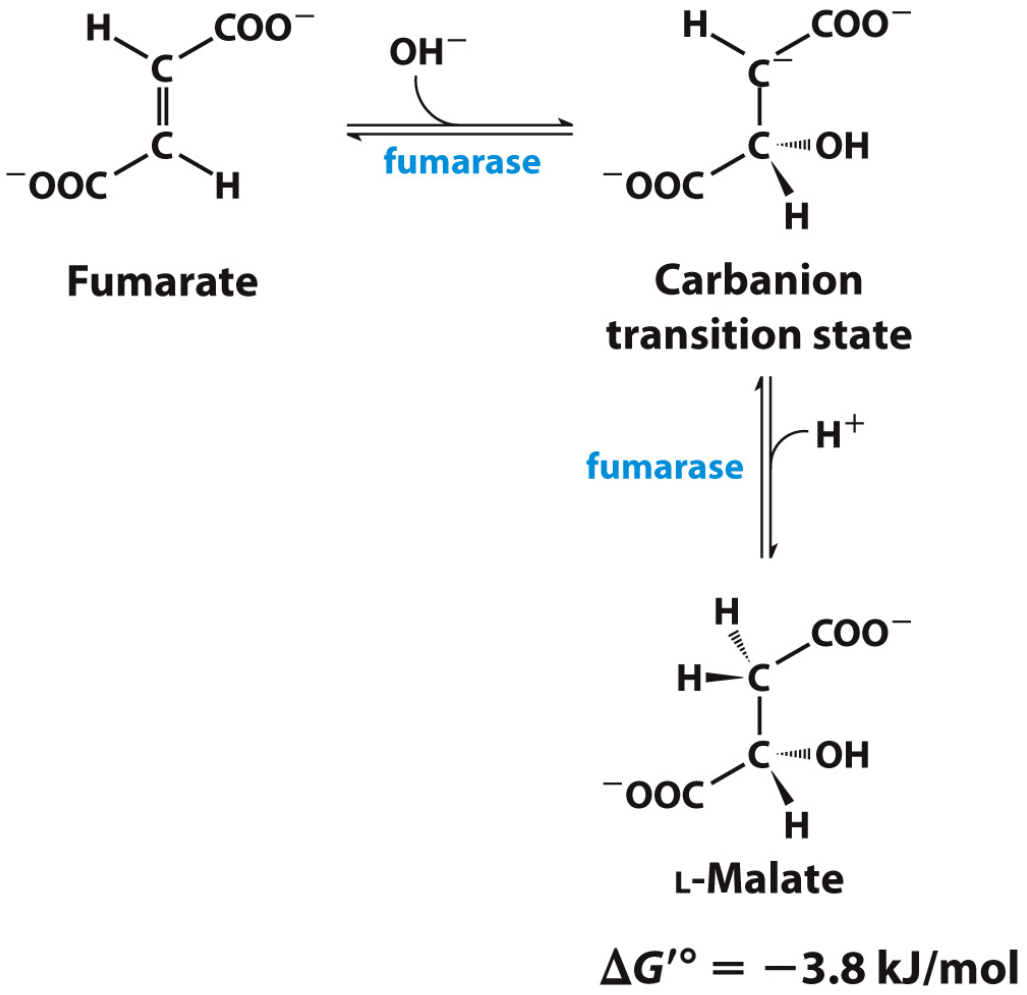
Succinate Dehydrogenase

- Bound to mitochondrial inner membrane
  - acts as Complex II in the electron-transport chain
- Reduction of the alkane to alkene requires $FADH_2$.
  - Reduction potential of carbon-hydrogen bond is too low for production of NADH.
- FAD is covalently bound, unusual
- Near equilibrium/reversible
  - product concentration kept low to pull forward

Strongly inhibited by malonate



Hydration Across a Double Bond: Fumarase



Fumarate

Carbanion transition state

L-Malate

$\Delta G'^{\circ} = -3.8$ kJ/mol

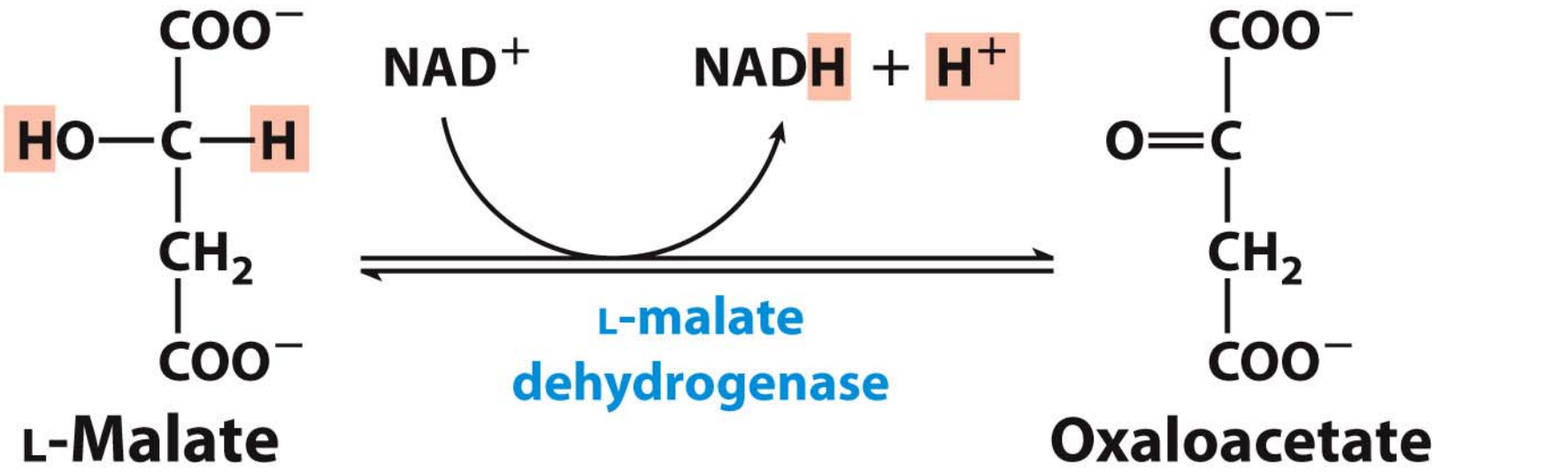
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Fumarase

- Stereospecific
  - Addition of water is always trans and forms l-malate.
  - OH^- adds to fumarate… then H^+ adds to the carbanion.
  - Cannot distinguish between inner carbons, so either can gain –OH

- Slightly thermodynamically favorable/reversible
  - product concentration kept low to pull reaction forward

Oxidation of Alcohol to a Ketone and Regeneration of Oxaloacetate by Malate Dehydrogenase



$\Delta G'^{\circ} = 29.7$ kJ/mol

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Malate Dehydrogenase

- Final step of the cycle
- Regenerates oxaloacetate for citrate synthase
- Highly thermodynamically **UN**favorable/reversible
  - oxaloacetate concentration kept VERY low by citrate synthase
    - pulls the reaction forward

One Turn of the Citric Acid Cycle

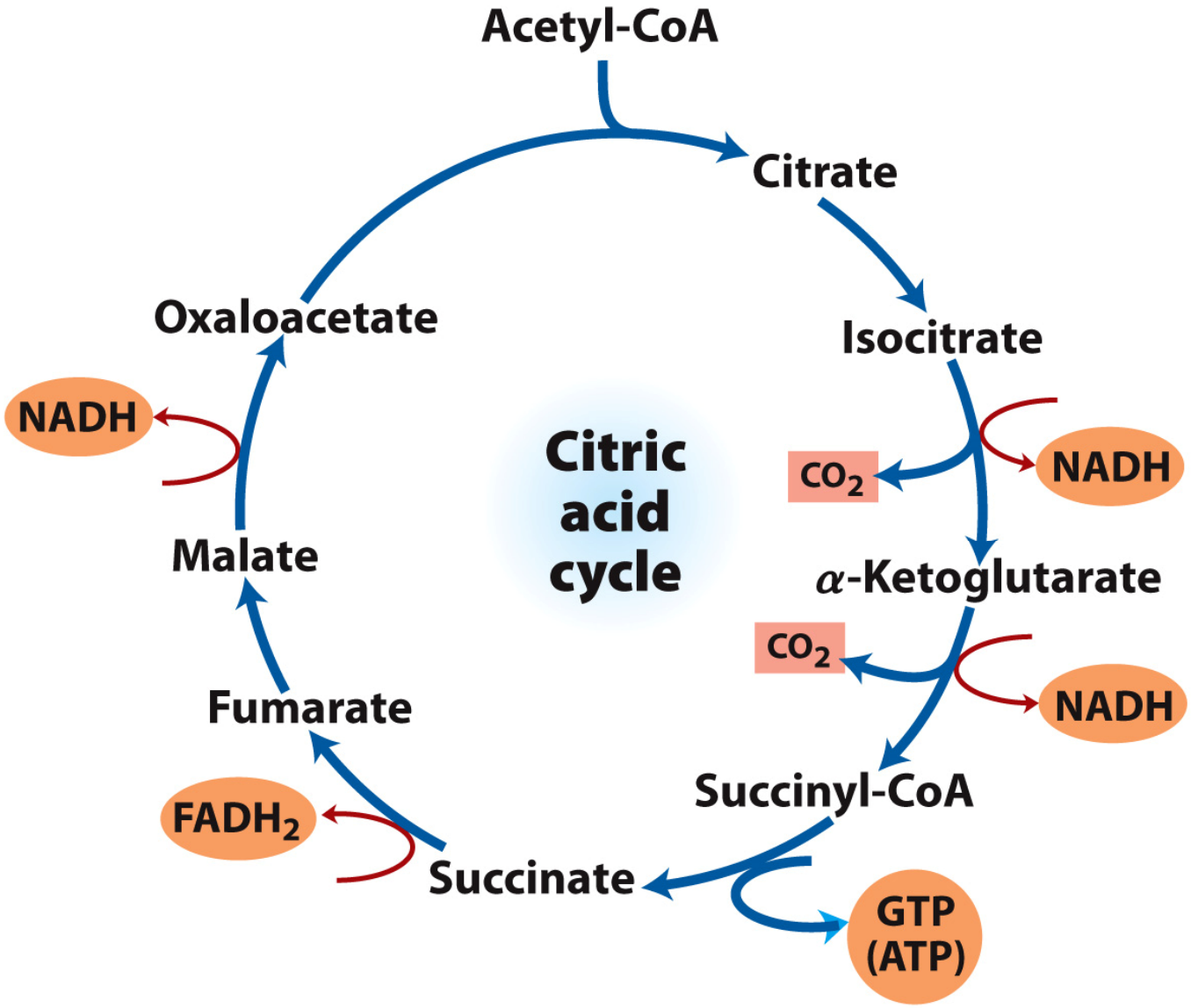


Figure 16-14
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Net Result of the Citric Acid Cycle

$$\text{Acetyl-CoA} + 3NAD^+ + FAD + GDP + P_i + 2\,H_2O \rightarrow$$

$$2CO_2 + 3NADH + FADH_2 + GTP + CoA + 3H^+$$

- Net oxidation of two carbons to CO_2
  - equivalent to two carbons of acetyl-CoA
  - but NOT the exact same carbons
- Energy captured by electron transfer to NADH and $FADH_2$
- Generates 1 GTP, which can be converted to ATP
- Completion of cycle

Direct and Indirect ATP Yield

TABLE 16-1 **Stoichiometry of Coenzyme Reduction and ATP Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Complex Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation**

Reaction	Number of ATP or reduced coenzyme directly formed	Number of ATP ultimately formed[a]
Glucose → glucose 6-phosphate	–1 ATP	–1
Fructose 6-phosphate → fructose 1,6-bisphosphate	–1 ATP	–1
2 Glyceraldehyde 3-phosphate → 2 1,3-bisphosphoglycerate	2 NADH	3 or 5[b]
2 1,3-Bisphosphoglycerate → 2 3-phosphoglycerate	2 ATP	2
2 Phosphoenolpyruvate → 2 pyruvate	2 ATP	2
2 Pyruvate → 2 acetyl-CoA	2 NADH	5
2 Isocitrate → 2 α-ketoglutarate	2 NADH	5
2 α-Ketoglutarate → 2 succinyl-CoA	2 NADH	5
2 Succinyl-CoA → 2 succinate	2 ATP (or 2 GTP)	2
2 Succinate → 2 fumarate	2 $FADH_2$	3
2 Malate → 2 oxaloacetate	2 NADH	5
Total		30-32

[a]This is calculated as 2.5 ATP per NADH and 1.5 ATP per $FADH_2$. A negative value indicates consumption.
[b]This number is either 3 or 5, depending on the mechanism used to shuttle NADH equivalents from the cytosol to the mitochondrial matrix; see Figures 19-30 and 19-31.

Regulation of the Citric Acid Cycle

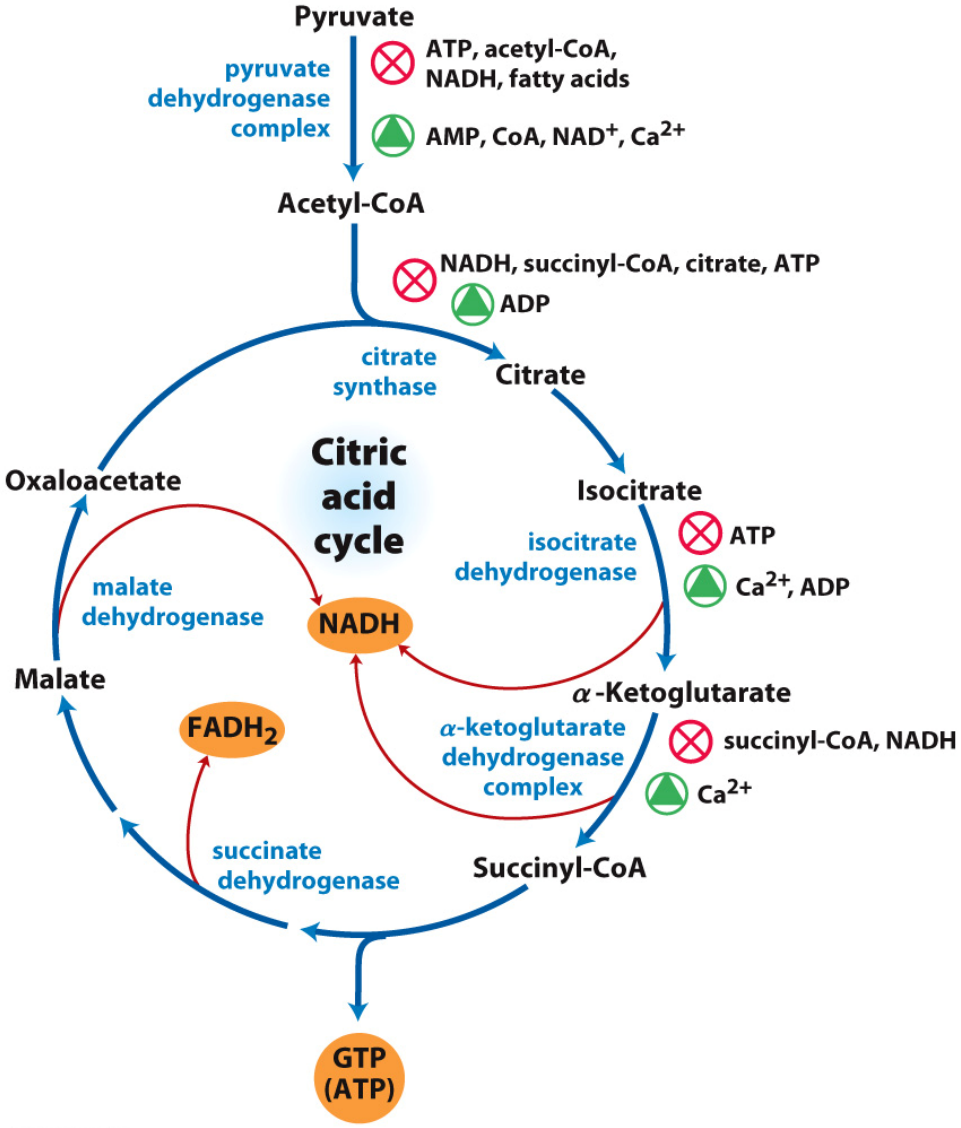


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Regulation of the Citric Acid Cycle

- Regulated at highly thermodynamically favorable and irreversible steps
  - PDH, citrate synthase, IDH, and KDH

- General regulatory mechanism
  - activated by substrate availability
  - inhibited by product accumulation
  - Overall products of the pathway are NADH and ATP.
    - affect all regulated enzymes in the cycle
    - inhibitors: NADH and ATP
    - activators: NAD+ and AMP

Regulation of Pyruvate Dehydrogenase

- Also regulated by reversible phosphorylation of E1
  - phosphorylation: inactive
  - dephosphorylation: active
- PDH kinase and PDH phosphorylase are part of mammalian PDH complex.
  - Kinase is activated by ATP.
    - high ATP → phosphorylated PDH → less acetyl-CoA
    - low ATP → kinase is less active and phosphorylase removes phosphate from PDH → more acetyl-CoA

Regulation of PDH is somewhat similar to regulation of the glycogen synthase/glycogen phosphorylase complex.

Additional Regulatory Mechanisms

- Citrate synthase is also inhibited by succinyl-CoA.
  - α-Ketoglutarate is an important branch point for amino acid metabolism.
  - Succinyl-CoA communicates flow at this branch point to the start of the cycle.

- Regulation of isocitrate dehydrogenase controls citrate levels.
  - Aconitase is reversible.
  - Inhibition of IDH leads to accumulation of isocitrate and reverses acconitase.
  - Accumulated citrate leaves mitochondria and inhibits phosphofructokinase in glycolysis.

CAC Intermediates Are Amphibolic

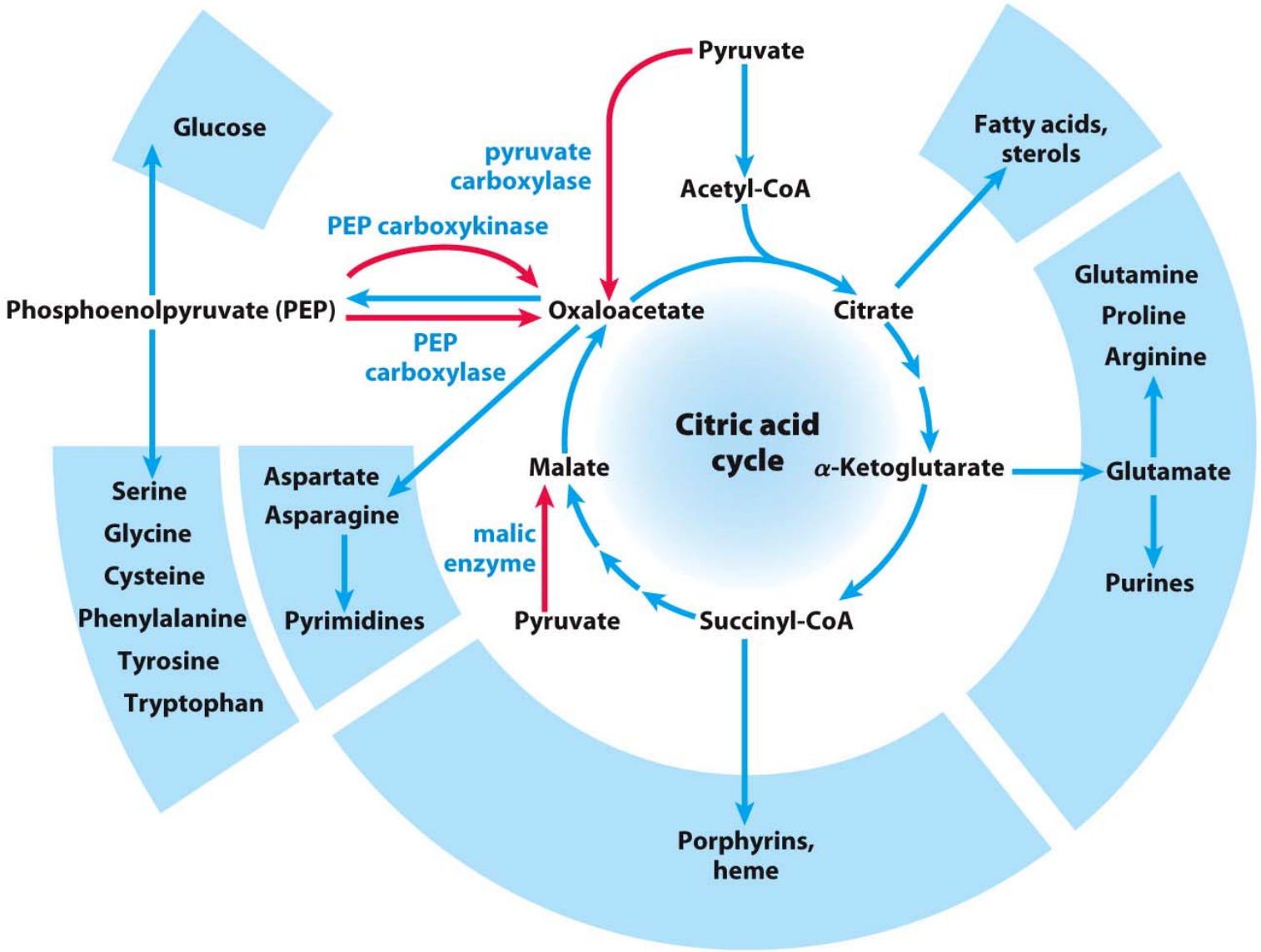


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Anaplerotic Reactions

- Intermediates in the citric acid cycle can be used in biosynthetic pathways (removed from cycle).
- Must replenish the intermediates in order for the cycle and central metabolic pathway to continue
- 4-carbon intermediates are formed by carboxylation of 3-carbon precursors.

TABLE 16-2 Anaplerotic Reactions

Reaction	Tissue(s)/organism(s)
$\text{Pyruvate} + HCO_3^- + \text{ATP} \xrightleftharpoons{\text{pyruvate carboxylase}} \text{oxaloacetate} + \text{ADP} + P_i$	Liver, kidney
$\text{Phosphoenolpyruvate} + CO_2 + \text{GDP} \xrightleftharpoons{\text{PEP carboxykinase}} \text{oxaloacetate} + \text{GTP}$	Heart, skeletal muscle
$\text{Phosphoenolpyruvate} + HCO_3^- \xrightleftharpoons{\text{PEP carboxylase}} \text{oxaloacetate} + P_i$	Higher plants, yeast, bacteria
$\text{Pyruvate} + HCO_3^- + \text{NAD(P)H} \xrightleftharpoons{\text{malic enzyme}} \text{malate} + \text{NAD(P)}^+$	Widely distributed in eukaryotes and bacteria

Table 16-2
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